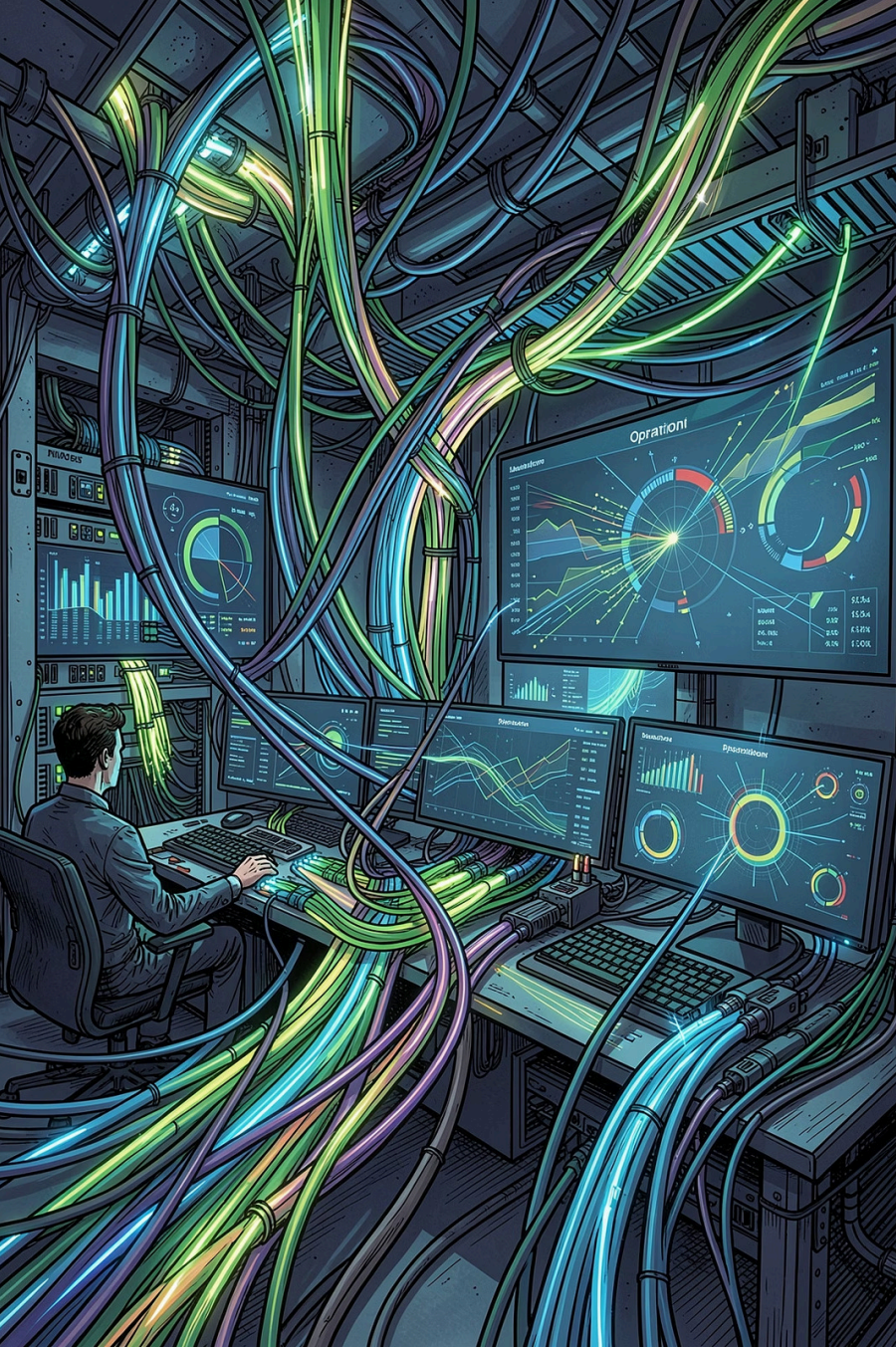




Fiber Field Operations Management System

A production-grade platform designed from real-world field observations – streamlining task dispatch, team coordination, and live execution tracking for fiber installation companies.

SYSTEM ARCHITECTURE

GRADUATION → ENTERPRISE

The Problem

Manual Operations Are Breaking Down



Supervisors receive daily job requests as PDFs or internal search results, then manually filter, assign, and track everything – with no unified system. This creates delays, errors, and lost visibility.

Fragmented Task Sources

PDFs, internal system searches, and manual inputs with no unified model

No Real-Time Visibility

Supervisors track progress manually with zero live field updates

Daily Dispatch Chaos

Teams rebuilt from scratch every day with spreadsheets and phone calls

System Overview

Two Platforms. One Operational Brain.



Admin Dispatch Dashboard

Web-based supervisor interface for importing tasks, building daily teams, assigning work, and monitoring live field execution.



Technician Mobile App

Field-facing application where technicians log in, see their team and tasks, and execute each step of the installation or repair workflow.

Core Goals

- Centralize all task sources into one unified model
- Enable dynamic daily dispatch team creation
- Deliver real-time synchronization between field and HQ
- Provide full task lifecycle visibility from import to review

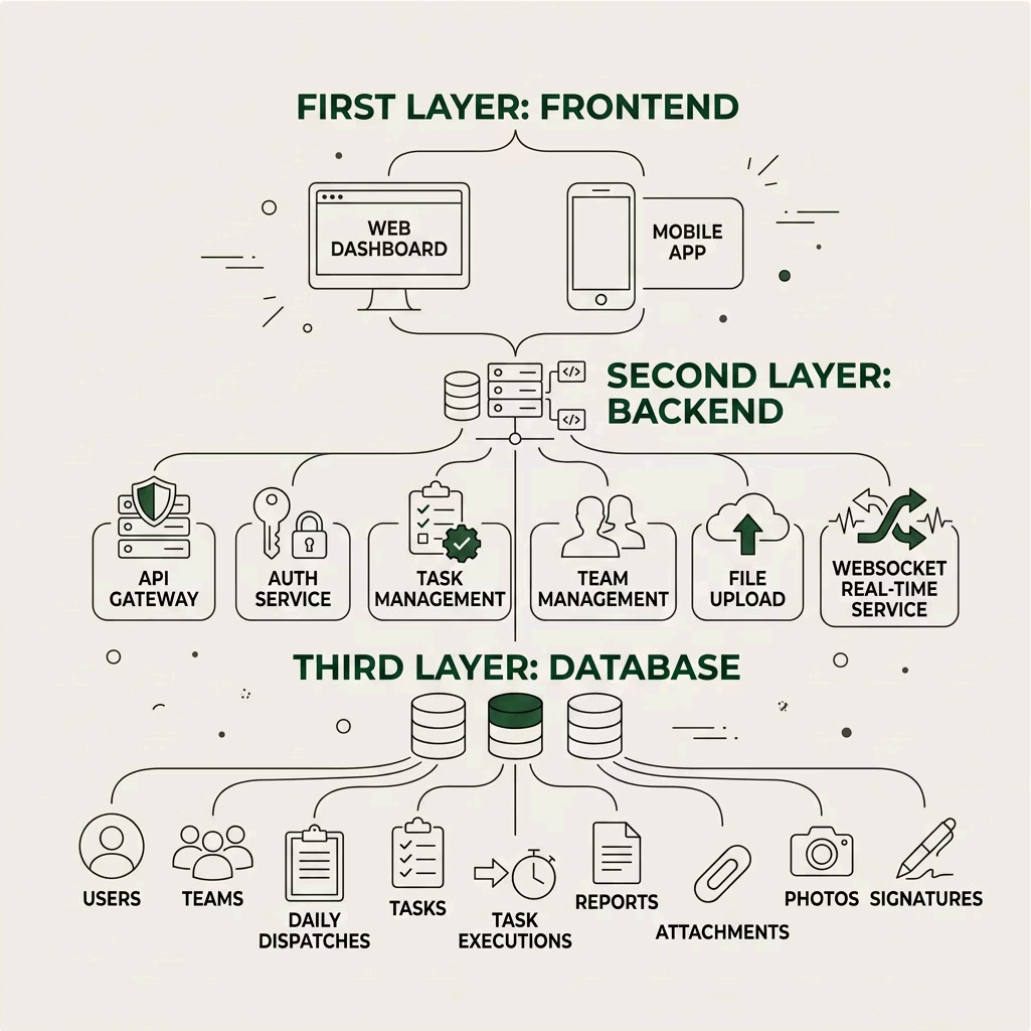
```
graph TD; DB[Database] --> UTM((Unified Task Model)); SRC[Source] --> UTM; UTM --> TPM[Task Parser & Mapper]; TPM --> UTM; UTM --> UTM_MODEL[Unified Task Model]; UTM_MODEL --> UTM;
```

The diagram illustrates the Unified Task Model architecture. It features a central dark green circle labeled "Unified Task Model". Surrounding this central circle are four light green rounded rectangular boxes, each representing a component of the system:

- Database**: Feeds dashboard and mobile app.
- Source**: PDF upload, API search, manual input.
- Task Parser & Mapper**: Extracts fields and maps to schema.
- Unified Task Model**: Normalizes every task to one structure.

Made with **GAMMA**

Full-Stack Technical Blueprint



External Data Sources

PDF Import

Upload, parse, and extract task IDs, addresses, POP, and customer data

Internal API Search

Search by task identifier (e.g. `TLC_00000000313996`) and fetch live data

Manual Creation

Supervisor-created tasks entered directly into the dashboard

i All sources pass through the Task Parser before reaching the database – never directly.

Daily Dispatch — Built for Speed and Flexibility



Dynamic Team Creation

Teams (Alpha, Beta, Gamma...) are built fresh each day based on task volume, regions, and availability



Drag & Drop Assignment

Assign workers and tasks visually — drag technicians into teams, drag tasks to assignments



POP / Region Grouping

Tasks are automatically grouped by POP and geographic region to optimize routing

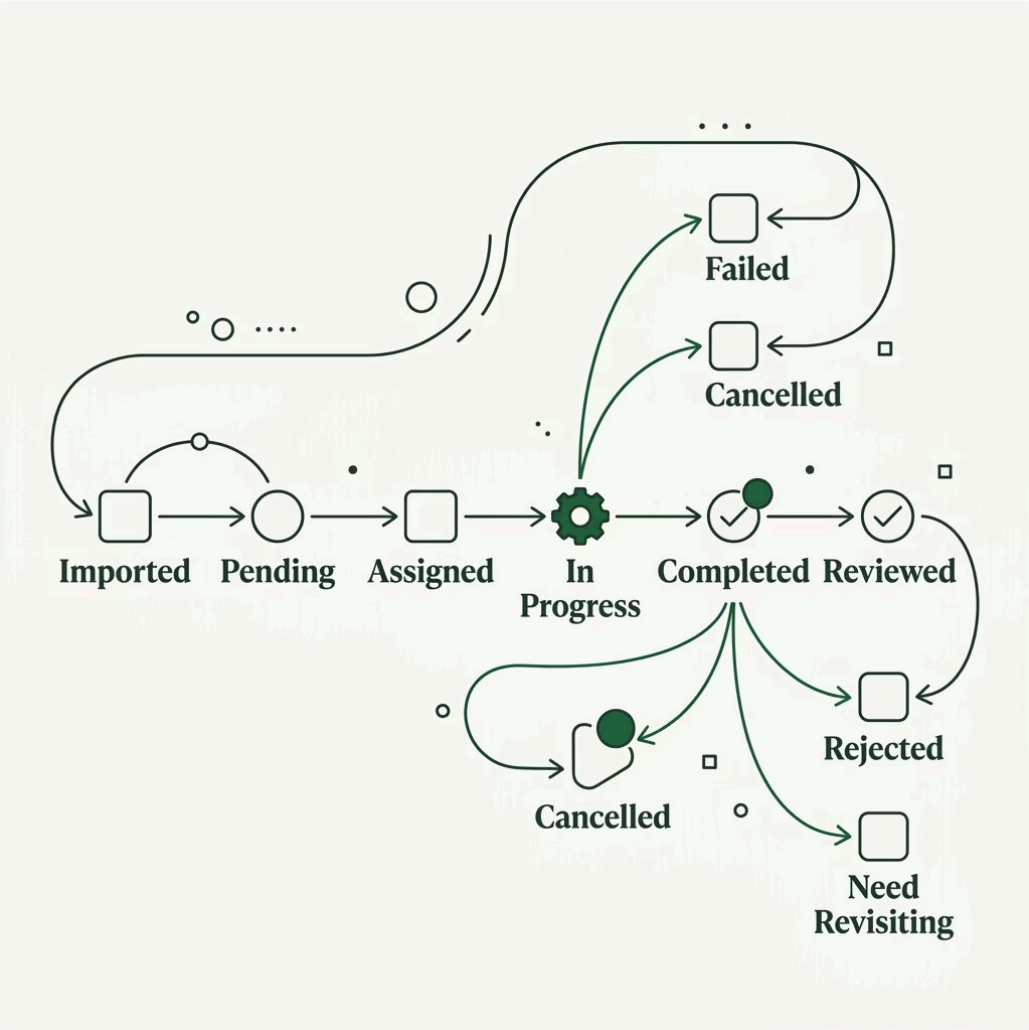


Live Field Monitoring

Real-time status updates from technicians flow directly into the supervisor's view



From Import to Completion — Every State Tracked



State Descriptions

Imported → Pending

Task ingested from any source and normalized; awaiting supervisor assignment

Assigned → In Progress

Technician has been dispatched and has checked in at the job site

Completed → Reviewed

All steps executed, verbally signed, supervisor confirms close-out

Exception States

Failed, Cancelled, Rejected, or Need Revisiting trigger re-dispatch logic

Technician Execution Workflow



Once a technician logs in, the app detects their assigned team and surfaces that day's tasks. Each job follows a structured step-by-step execution flow:

01

Check-In

GPS-confirmed arrival at job site

02

Adduzione TIM / Collegamento FO

Fiber connection steps with yes/no validations and photo uploads

03

Collaudo & Attivazione

Testing and activation with text inputs and notes

04

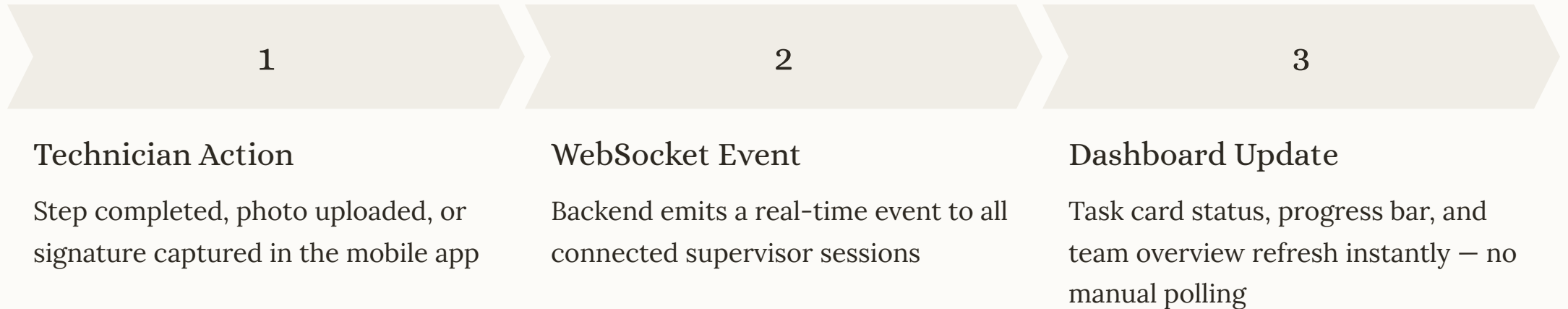
Esito, Signature & Verbale

Customer signature capture and final report generation

Real-Time Sync

Field and HQ — Always in Sync

WebSocket connections keep supervisor dashboards live as technicians update job status in the field. No refresh. No lag. No phone calls.



- ✔ Live sync events include: task started, photo uploaded, customer signed, installation completed, repair failed — all surfaced in real time to the supervisor.

Built to Grow — From Graduation Project to Production Platform

1

Architecture Finalized

Unified task model, dispatch logic, and real-time sync fully designed and documented

2

Core Services Scoped

Task, Team, File, Auth, and WebSocket services ready for implementation sprint planning

3

API Integration Ready

Parser/mapper layer supports future live API feeds from the company's internal system with zero rework

4

Production-Grade Foundation

Schema, lifecycle states, and dispatch model designed for scale — not just graduation

